

International Accounting Software

Complete SQL Database Schema & Security Architecture Reference

ASP.NET Core (Web API) | SQL Server | React (JavaScript)

This document defines a complete, modular relational schema for a multi-company, multi-currency, multi-branch accounting system intended to meet the needs of any type of business, from a single retail shop to an international trading group. It is designed to be implemented incrementally: build and run Module 1-3 first (core + security + General Ledger), confirm it posts balanced journals correctly, then add the remaining modules one at a time using the same conventions.

Conventions Used Throughout This Schema

Standard columns on every table (not repeated below): Id (BIGINT IDENTITY, PK) | CompanyId (BIGINT, FK → Companies.Id, tenant isolation) | CreatedAt (DATETIME2) | CreatedBy (BIGINT, FK → Users.Id) | ModifiedAt (DATETIME2, NULL) | ModifiedBy (BIGINT, NULL) | IsDeleted (BIT, DEFAULT 0, soft delete) | RowVersion (ROWVERSION, optimistic concurrency). Tables that are global master/lookup data (e.g. Currencies, Countries, Permissions) omit CompanyId.

- **Naming:** PascalCase for tables and columns; tables named in the plural (Vendors, JournalEntries).
- **Primary keys:** BIGINT IDENTITY(1,1). Use UNIQUEIDENTIFIER instead only if you need globally-unique keys generated client-side or across distributed databases.
- **Money columns:** DECIMAL(18,4) everywhere - never FLOAT/REAL for currency.
- **Dates/times:** DATE for calendar dates, DATETIME2(3) for timestamps, always stored in UTC.
- **Soft delete:** business/master data uses IsDeleted; posted transactional data (JournalEntries, invoices once Posted) is never deleted at all - only reversed or voided, to preserve the audit trail.
- **Multi-tenancy:** every business table carries CompanyId; enforce isolation with an EF Core global query filter AND a database-level check, not application logic alone.
- **Referential integrity on transactional tables:** use FK ON DELETE NO ACTION (never CASCADE) so a mistaken delete can never ripple through financial history.
- **Double-entry rule:** enforced at the application layer inside the same DB transaction that inserts JournalEntry + its JournalEntryLines - reject if $\text{SUM}(\text{Debit}) \neq \text{SUM}(\text{Credit})$.

Recommended Build & Rollout Phases

Build in this order so that at every stage you have something you can actually run and test end-to-end:

Phase 1 - Foundation

Module 1 (Core Setup) + Module 2 (Security & Access) + Module 3 (Chart of Accounts & General Ledger). Goal: log in, create a company, post a manual balanced journal, see it on a trial balance.

Phase 2 - Trading Cycle

Module 6 (Accounts Payable) + Module 7 (Accounts Receivable) + Module 8 (Banking & Cash). Goal: raise a sales invoice, receive payment, enter a vendor bill, pay it, reconcile the bank.

Phase 3 - Compliance & Control

Module 5 (Tax) + Module 4 (Dimensions) + approval workflow tables in Module 2. Goal: correct tax on every invoice, cost-center/project reporting, and amounts above a threshold requiring sign-off.

Phase 4 - Assets & Planning

Module 9 (Fixed Assets) + Module 10 (Budgeting) + Module 11 (Basic Inventory/COGS).

Phase 5 - Extended & Reporting

Module 12 (Payroll) + Module 13 (Financial Reporting & Consolidation) + Module 14 (Notifications).

1. Core Setup & Multi-Tenancy

Foundation tables every other module depends on: legal entities, branches, fiscal calendar, currencies, document numbering and file attachments. Designed so the same database can host multiple companies/tenants safely (every business table carries a CompanyId).

Companies

One row per legal entity / tenant using the system.

Column	Data Type	Key / Constraint	Description
Name	NVARCHAR(200)	NOT NULL	Trading name
LegalName	NVARCHAR(250)	NULL	Registered legal name
RegistrationNumber	NVARCHAR(100)	NULL	Company/trade license number
TaxIdentificationNumber	NVARCHAR(100)	NULL	TIN / VAT registration number
CountryCode	CHAR(2)	NOT NULL, FK->Countries.Code	ISO country of incorporation
BaseCurrencyId	BIGINT	NOT NULL, FK->Currencies.Id	Functional/reporting currency
FiscalYearStartMonth	TINYINT	NOT NULL	1-12, start month of fiscal year
Address	NVARCHAR(400)	NULL	Registered address
Phone/Email	NVARCHAR(100)	NULL	Primary contact
LogoPath	NVARCHAR(300)	NULL	Path/URL to logo for printed documents
IsActive	BIT	NOT NULL DEFAULT 1	Soft enable/disable of tenant

Branches

Physical offices/warehouses/outlets under a company.

Column	Data Type	Key / Constraint	Description
Name	NVARCHAR(150)	NOT NULL	Branch name
Code	NVARCHAR(20)	NOT NULL, UNIQUE per Company	Short code used in voucher numbers
Address	NVARCHAR(400)	NULL	Branch address
IsHeadOffice	BIT	NOT NULL DEFAULT 0	Marks the head-office branch
IsActive	BIT	NOT NULL DEFAULT 1	

Countries / Currencies

Global master lists (no CompanyId).

Column	Data Type	Key / Constraint	Description
Countries.Code	CHAR(2)	PK (ISO-3166)	Country code
Countries.Name	NVARCHAR(100)	NOT NULL	Country name
Currencies.Code	CHAR(3)	PK (ISO-4217)	e.g. USD, EUR, BDT, AED
Currencies.Name	NVARCHAR(60)	NOT NULL	e.g. US Dollar
Currencies.Symbol	NVARCHAR(10)	NULL	e.g. \$, €, ■
Currencies.DecimalPlaces	TINYINT	NOT NULL DEFAULT 2	Minor unit precision (0-4)

ExchangeRates

Daily/point-in-time FX rates between currency pairs.

Column	Data Type	Key / Constraint	Description
FromCurrencyId	CHAR(3)	FK->Currencies.Code	Source currency
ToCurrencyId	CHAR(3)	FK->Currencies.Code	Target currency (usually company base)
RateDate	DATE	NOT NULL	Effective date
Rate	DECIMAL(18,8)	NOT NULL	Multiply From-amount by this to get To-amount
RateType	NVARCHAR(20)	NOT NULL DEFAULT 'Closing'	Spot / Average / Closing / Budget
Source	NVARCHAR(100)	NULL	Central bank / provider name

UNIQUE constraint on (FromCurrencyId, ToCurrencyId, RateDate, RateType).

FiscalYears / AccountingPeriods

Accounting calendar and period locking.

Column	Data Type	Key / Constraint	Description
FiscalYears.Name	NVARCHAR(20)	NOT NULL	e.g. FY2026-27
FiscalYears.StartDate/End Date	DATE	NOT NULL	Year boundaries
FiscalYears.IsClosed	BIT	NOT NULL DEFAULT 0	Blocks further posting once year-end run
AccountingPeriods.FiscalYearId	BIGINT	FK->FiscalYears.Id	Parent year
AccountingPeriods.PeriodNumber	TINYINT	NOT NULL	1-12 (or 13 for adjustment period)
AccountingPeriods.StartDate/EndDate	DATE	NOT NULL	Period boundaries
AccountingPeriods.IsClosed	BIT	NOT NULL DEFAULT 0	Period-level lock, checked before every post

NumberSequences

Centralised auto-numbering for every document type, per company/branch.

Column	Data Type	Key / Constraint	Description
BranchId	BIGINT	NULL, FK->Branches.Id	Branch-specific sequence, or company-wide if NULL
DocumentType	NVARCHAR(40)	NOT NULL	JournalVoucher, SalesInvoice, PurchaseBill, Receipt, Payment, PO, SO ...
Prefix	NVARCHAR(15)	NULL	e.g. 'INV-', 'JV-'
NextNumber	BIGINT	NOT NULL DEFAULT 1	Next value to issue (updated inside the posting transaction)
PaddingLength	TINYINT	NOT NULL DEFAULT 6	Zero-pad width, e.g. 000123
ResetFrequency	NVARCHAR(10)	NOT NULL DEFAULT 'Yearly'	Never / Yearly / Monthly
LastResetDate	DATE	NULL	Last time counter was reset

UNIQUE (CompanyId, BranchId, DocumentType). Guard with SQL SEQUENCE or row-lock to avoid duplicate numbers under concurrency.

Attachments

Generic polymorphic file store used by every module (invoices, journals, assets, HR docs...).

Column	Data Type	Key / Constraint	Description
EntityType	NVARCHAR(60)	NOT NULL	e.g. 'PurchaseInvoice', 'FixedAsset'
EntityId	BIGINT	NOT NULL	PK value of the referenced row
FileName	NVARCHAR(260)	NOT NULL	Original file name
FilePath	NVARCHAR(400)	NOT NULL	Storage path / blob URL
FileSizeBytes	BIGINT	NOT NULL	
ContentType	NVARCHAR(100)	NOT NULL	MIME type
UploadedBy	BIGINT	FK->Users.Id	
UploadedAt	DATETIME2	NOT NULL	

Index on (EntityType, EntityId). Never expose FilePath directly to the client; stream via an authorized API.

SystemSettings

Per-company key/value configuration (rounding rules, invoice footer text, default terms, etc.).

Column	Data Type	Key / Constraint	Description
SettingKey	NVARCHAR(100)	NOT NULL	e.g. 'DefaultPaymentTermsDays'
SettingValue	NVARCHAR(MAX)	NULL	Stored as text/JSON, parsed by application layer

2. Security, Users & Access Control

Authentication, role-based authorization, tenant/branch scoping, approval workflow and a full audit trail. This module is the backbone that keeps the accounting data trustworthy and tamper-evident.

Users

One row per login (a user may belong to more than one company via UserCompanyAccess).

Column	Data Type	Key / Constraint	Description
Username	NVARCHAR(60)	UNIQUE, NOT NULL	
Email	NVARCHAR(150)	UNIQUE, NOT NULL	
PasswordHash	NVARCHAR(255)	NOT NULL	Argon2id or bcrypt hash - never store plain text
PasswordSalt	NVARCHAR(128)	NOT NULL	Unique per-user salt (if algorithm needs it separately)
IsEmailVerified	BIT	NOT NULL DEFAULT 0	
PhoneNumber	NVARCHAR(30)	NULL	
IsTwoFactorEnabled	BIT	NOT NULL DEFAULT 0	
TwoFactorSecretEncrypted	NVARCHAR(255)	NULL	TOTP secret, encrypted at rest
IsActive / IsLocked	BIT	NOT NULL DEFAULT 1 / 0	Account enable + lockout flags
FailedLoginAttempts	INT	NOT NULL DEFAULT 0	Reset on success; trigger lock after N attempts
LastLoginAt	DATETIME2	NULL	
LastPasswordChangeAt	DATETIME2	NULL	Enforce periodic rotation policy if required
MustChangePassword	BIT	NOT NULL DEFAULT 0	Force reset on first login / after admin reset

Roles / Permissions

Role-Based Access Control (RBAC).

Column	Data Type	Key / Constraint	Description
Roles.Name	NVARCHAR(80)	NOT NULL	e.g. Admin, Accountant, Auditor, DataEntry, ApproverL1
Roles.IsSystemRole	BIT	NOT NULL DEFAULT 0	System roles cannot be deleted
Permissions.ModuleName	NVARCHAR(60)	NOT NULL	e.g. 'GeneralLedger', 'AccountsPayable'
Permissions.Action	NVARCHAR(30)	NOT NULL	View / Create / Edit / Delete / Approve / Post / Void
Permissions.Code	NVARCHAR(100)	UNIQUE	Machine key, e.g. 'AP.INVOICE.APPROVE'
RolePermissions	table	PK(RoleId, PermissionId)	Junction: which permissions a role grants
UserRoles	table	PK(UserId, RoleId)	Junction: which roles a user has (per company)

UserCompanyAccess

Restricts which company/branch a given user may sign into and act on.

Column	Data Type	Key / Constraint	Description
UserId	BIGINT	FK->Users.Id	
BranchId	BIGINT	NULL, FK->Branches.Id	NULL = access to all branches of the company
IsDefault	BIT	NOT NULL DEFAULT 0	Pre-selected company on login

RefreshTokens

JWT refresh-token rotation store (access tokens stay short-lived and stateless).

Column	Data Type	Key / Constraint	Description
UserId	BIGINT	FK->Users.Id	
TokenHash	NVARCHAR(255)	NOT NULL	Store a hash of the token, never the raw value
ExpiresAt	DATETIME2	NOT NULL	
CreatedByIp	NVARCHAR(45)	NULL	
RevokedAt	DATETIME2	NULL	
ReplacedByTokenHash	NVARCHAR(255)	NULL	Enables rotation-reuse-detection

LoginHistory

Every login attempt, for security monitoring.

Column	Data Type	Key / Constraint	Description
UserId	BIGINT	NULL, FK->Users.Id	NULL if username did not resolve to a user
AttemptedUsername	NVARCHAR(60)	NULL	
LoginAt	DATETIME2	NOT NULL	
IpAddress	NVARCHAR(45)	NOT NULL	
UserAgent	NVARCHAR(300)	NULL	
Success	BIT	NOT NULL	
FailureReason	NVARCHAR(100)	NULL	WrongPassword / Locked / Disabled / MfaFailed

AuditLogs

Immutable, append-only change log for every insert/update/delete/post/approve/reject on financial data.

Column	Data Type	Key / Constraint	Description
UserId	BIGINT	FK->Users.Id	Who performed the action
EntityName	NVARCHAR(80)	NOT NULL	Table/entity affected
EntityId	BIGINT	NOT NULL	Row affected
Action	NVARCHAR(20)	NOT NULL	Insert / Update / Delete / Post / Approve / Reject / Void
OldValuesJson	NVARCHAR(MAX)	NULL	Snapshot before change
NewValuesJson	NVARCHAR(MAX)	NULL	Snapshot after change
Timestamp	DATETIME2	NOT NULL	
IpAddress	NVARCHAR(45)	NULL	

Application must write here in the SAME database transaction as the business change (or via CDC/trigger) so the trail can never be bypassed. Grant the app's SQL login INSERT-only rights on this table.

Approval Workflow

Configurable multi-step approval before a document can be posted (e.g. invoices above a threshold).

Column	Data Type	Key / Constraint	Description
ApprovalWorkflows.EntityType	NVARCHAR(60)	NOT NULL	e.g. 'PurchaseInvoice'
ApprovalWorkflows.MinAmount/MaxAmount	DECIMAL(18,4)	NULL	Amount band the workflow applies to
ApprovalSteps.WorkflowId	BIGINT	FK	
ApprovalSteps.StepOrder	TINYINT	NOT NULL	
ApprovalSteps.ApproverRoleId	BIGINT	NULL, FK->Roles.Id	Anyone holding this role can act
ApprovalRequests.EntityId/EntityType	BIGINT/NVARCHAR	NOT NULL	Document under review
ApprovalRequests.Status	NVARCHAR(20)	NOT NULL	Pending / Approved / Rejected / Cancelled
ApprovalActions.ActionBy/Action/Comments	-	-	History of each approve/reject decision

3. Chart of Accounts & General Ledger (Core Double-Entry Engine)

The heart of the system. Every transaction in every other module ultimately posts a balanced JournalEntry here. Designed to support unlimited account hierarchy, multi-currency lines and cost/project dimensions.

AccountTypes

Fixed master list (seeded once): Asset, Liability, Equity, Revenue, Expense.

Column	Data Type	Key / Constraint	Description
Name	NVARCHAR(30)	UNIQUE	Asset / Liability / Equity / Revenue / Expense
NormalBalance	CHAR(6)	NOT NULL	'Debit' or 'Credit'

AccountGroups

Hierarchical grouping used for reporting (e.g. Current Assets > Cash & Bank).

Column	Data Type	Key / Constraint	Description
ParentGroupId	BIGINT	NULL, FK->AccountGroups.Id	Self-referencing tree
Name	NVARCHAR(120)	NOT NULL	
AccountTypeId	BIGINT	FK->AccountTypes.Id	
DisplayOrder	INT	NOT NULL DEFAULT 0	Sort order in reports

ChartOfAccounts

Every ledger account (also called the GL Account master).

Column	Data Type	Key / Constraint	Description
AccountCode	NVARCHAR(20)	NOT NULL, UNIQUE per Company	e.g. 1010-01
AccountName	NVARCHAR(150)	NOT NULL	
ParentAccountId	BIGINT	NULL, FK->ChartOfAccounts.Id	For sub-ledger accounts
AccountGroupId	BIGINT	FK->AccountGroups.Id	
AccountTypeId	BIGINT	FK->AccountTypes.Id	Determines normal balance
CurrencyId	CHAR(3)	NULL, FK->Currencies.Code	NULL = base currency account
IsControlAccount	BIT	NOT NULL DEFAULT 0	True for AR/AP/Bank control accounts
ControlAccountFor	NVARCHAR(20)	NULL	Customer / Vendor / Bank / Employee / Inventory
AllowManualEntry	BIT	NOT NULL DEFAULT 1	False for system-only control accounts
OpeningBalance / OpeningBalanceDate	DECIMAL(18,4)/DATE	NULL	Migration opening balances
IsActive	BIT	NOT NULL DEFAULT 1	

CHECK: an account cannot be its own ancestor (enforce in application/trigger to avoid circular hierarchy).

JournalEntries

Header of every accounting transaction (manual journal, or auto-generated from AP/AR/Bank/Assets/Payroll).

Column	Data Type	Key / Constraint	Description
BranchId	BIGINT	FK->Branches.Id	
AccountingPeriodId	BIGINT	FK->AccountingPeriods.Id	Blocks posting into a closed period
VoucherNumber	NVARCHAR(30)	UNIQUE per Company	From NumberSequences
VoucherDate	DATE	NOT NULL	
VoucherType	NVARCHAR(20)	NOT NULL	General / Sales / Purchase / Receipt / Payment / Contra / Adjustment / Depreciation / FxRevaluation
SourceModule / SourceDocumentId	NVARCHAR(40)/BIGINT	NULL	Traceability back to the originating AP/AR/Asset document
Reference	NVARCHAR(100)	NULL	External reference
Narration	NVARCHAR(400)	NULL	
CurrencyId	CHAR(3)	FK->Currencies.Code	Transaction currency
ExchangeRateToBase	DECIMAL(18,8)	NOT NULL DEFAULT 1	
Status	NVARCHAR(20)	NOT NULL	Draft / PendingApproval / Posted / Rejected / Reversed
PostedAt / PostedBy	DATETIME2/BIGINT	NULL	Set only on posting - posted entries become immutable
ReversalOfEntryId	BIGINT	NULL, FK->JournalEntries.Id	Link when this entry reverses another

JournalEntryLines

Debit/credit lines of a journal entry. Sum(Debit) must equal Sum(Credit) per entry.

Column	Data Type	Key / Constraint	Description
JournalEntryId	BIGINT	FK->JournalEntries.Id	
LineNumber	INT	NOT NULL	
AccountId	BIGINT	FK->ChartOfAccounts.Id	
CostCenterId / ProjectId / DepartmentId	BIGINT	NULL, FK	Analytical dimensions
DebitAmount / CreditAmount	DECIMAL(18,4)	NOT NULL DEFAULT 0	CHECK: exactly one of the two is non-zero
BaseCurrencyDebit / BaseCurrencyCredit	DECIMAL(18,4)	NOT NULL	Converted using entry's exchange rate
Description	NVARCHAR(300)	NULL	
PartyType / PartyId	NVARCHAR(20)/BIGINT	NULL	Customer / Vendor / Employee sub-ledger link
TaxCodeId	BIGINT	NULL, FK->TaxCodes.Id	
ReconciliationStatus	NVARCHAR(15)	NOT NULL DEFAULT 'Open'	Open / Reconciled / Cleared (for bank & AR/AP matching)

Trigger/application rule: block INSERT/UPDATE/DELETE on lines whose parent JournalEntries.Status = 'Posted'. Corrections are made via a new reversing entry, never by editing history.

RecurringJournalTemplates

Templates that auto-generate journals on a schedule (rent, depreciation, accruals).

Column	Data Type	Key / Constraint	Description
Name	NVARCHAR(120)	NOT NULL	
Frequency	NVARCHAR(15)	NOT NULL	Daily / Weekly / Monthly / Quarterly / Yearly
NextRunDate	DATE	NOT NULL	
TemplateLinesJson	NVARCHAR(MAX)	NOT NULL	Line template (accounts, %/fixed amounts)
IsActive	BIT	NOT NULL DEFAULT 1	

4. Analytical Dimensions

Optional slicing dimensions attached to journal lines and sub-ledger transactions for management reporting (profit by department, by project, by cost center) without multiplying the chart of accounts.

Departments

Organisational departments (can be hierarchical).

Column	Data Type	Key / Constraint	Description
Name	NVARCHAR(120)	NOT NULL	
Code	NVARCHAR(20)	UNIQUE per Company	
ParentDepartmentId	BIGINT	NULL, FK->Departments.Id	

CostCenters

Cost-tracking units, optionally tied to a department.

Column	Data Type	Key / Constraint	Description
Name	NVARCHAR(120)	NOT NULL	
Code	NVARCHAR(20)	UNIQUE per Company	
DepartmentId	BIGINT	NULL, FK->Departments.Id	

Projects

Job/contract costing across revenue and expense accounts.

Column	Data Type	Key / Constraint	Description
Name	NVARCHAR(150)	NOT NULL	
Code	NVARCHAR(20)	UNIQUE per Company	
CustomerId	BIGINT	NULL, FK->Customers.Id	For billable projects
StartDate / EndDate	DATE	NULL	
BudgetAmount	DECIMAL(18,4)	NULL	
Status	NVARCHAR(20)	NOT NULL DEFAULT 'Active'	Active / OnHold / Closed

5. Multi-Jurisdiction Tax Management

Supports VAT/GST/Sales-Tax/Withholding-Tax across multiple countries and rates simultaneously, so the same install can serve businesses trading internationally.

TaxJurisdictions

Country/state/region a tax rule belongs to.

Column	Data Type	Key / Constraint	Description
CountryCode	CHAR(2)	FK->Countries.Code	
Region	NVARCHAR(100)	NULL	State/Province, if tax varies sub-nationally
Name	NVARCHAR(120)	NOT NULL	

TaxTypes

VAT, GST, Sales Tax, Withholding Tax, Excise Duty, etc.

Column	Data Type	Key / Constraint	Description
Name	NVARCHAR(40)	NOT NULL	
Description	NVARCHAR(200)	NULL	

TaxCodes

The actual rate(s) applied to transactions.

Column	Data Type	Key / Constraint	Description
TaxTypeId	BIGINT	FK->TaxTypes.Id	
JurisdictionId	BIGINT	FK->TaxJurisdictions.Id	
Code	NVARCHAR(20)	UNIQUE per Company	e.g. 'VAT15', 'WHT-5'
Name	NVARCHAR(80)	NOT NULL	
Rate	DECIMAL(7,4)	NOT NULL	Percentage, e.g. 15.0000
EffectiveFrom / EffectiveTo	DATE	NULL	Supports rate changes over time
IsCompound	BIT	NOT NULL DEFAULT 0	Tax-on-tax handling
IsRecoverable	BIT	NOT NULL DEFAULT 1	Input tax reclaimable vs a cost
PayableAccountId / ReceivableAccountId	BIGINT	FK->ChartOfAccounts.Id	GL postings for output/input tax

TaxTransactions

Line-level tax breakdown captured against source documents, feeding the tax return report.

Column	Data Type	Key / Constraint	Description
SourceDocumentType / SourceDocumentId	NVARCHAR(40)/BIGINT	NOT NULL	SalesInvoice / PurchaseInvoice / JournalEntry line
TaxCodeId	BIGINT	FK->TaxCodes.Id	
TaxableAmount	DECIMAL(18,4)	NOT NULL	
TaxAmount	DECIMAL(18,4)	NOT NULL	
JournalEntryId	BIGINT	FK->JournalEntries.Id	

6. Accounts Payable (Purchases & Vendor Payments)

Vendor master, purchase ordering, bills, and payment allocation with full multi-currency and withholding-tax support.

Vendors

Supplier master, doubles as the AP sub-ledger key.

Column	Data Type	Key / Constraint	Description
VendorCode	NVARCHAR(20)	UNIQUE per Company	
Name / LegalName	NVARCHAR(200)	NOT NULL	
TaxIdentificationNumber	NVARCHAR(100)	NULL	
CountryCode	CHAR(2)	FK->Countries.Code	
DefaultCurrencyId	CHAR(3)	FK->Currencies.Code	
PaymentTermsDays	INT	NOT NULL DEFAULT 30	
ControlAccountId	BIGINT	FK->ChartOfAccounts.Id	The AP control account this vendor rolls up to
DefaultExpenseAccountId	BIGINT	NULL, FK	Pre-fill on new bills
CreditLimit	DECIMAL(18,4)	NULL	
IsActive	BIT	NOT NULL DEFAULT 1	

VendorContacts / VendorBankAccounts

Supporting contact & remittance details.

Column	Data Type	Key / Constraint	Description
VendorContacts.Name/Designation/Email/Phone	-	-	One vendor can have many contacts
VendorContacts.IsPrimary	BIT	DEFAULT 0	
VendorBankAccounts.BankName/AccountNumber	NVARCHAR	NOT NULL	
VendorBankAccounts.IBAN / SwiftCode	NVARCHAR(50)	NULL	For international wire payments
VendorBankAccounts.CurrencyId	CHAR(3)	FK	

PurchaseOrders / PurchaseOrderLines

Optional PO stage before a bill is received.

Column	Data Type	Key / Constraint	Description
PurchaseOrders.VendorId	BIGINT	FK->Vendors.Id	
PurchaseOrders.PONumber/OrderDate/ExpectedDate	-	NOT NULL	
PurchaseOrders.Status	NVARCHAR(20)	NOT NULL	Draft/Sent/PartiallyReceived/Closed/Cancelled
PurchaseOrderLines.ItemId/Description/Quantity/UnitPrice	-	-	
PurchaseOrderLines.TaxCodeId / AccountId	BIGINT	FK	

PurchaseInvoices

Vendor bills - the core AP transaction, always posts a balanced JournalEntry.

Column	Data Type	Key / Constraint	Description
BranchId	BIGINT	FK->Branches.Id	
VendorId	BIGINT	FK->Vendors.Id	
PurchaseOrderId	BIGINT	NULL, FK->PurchaseOrders.Id	
InvoiceNumber	NVARCHAR(30)	UNIQUE per Company	Internal number
VendorInvoiceReference	NVARCHAR(60)	NULL	Vendor's own invoice number - prevents duplicate booking
InvoiceDate / DueDate	DATE	NOT NULL	
CurrencyId / ExchangeRateToBase	CHAR(3)/DECIMAL	NOT NULL	
SubTotal / TaxTotal / TotalAmount	DECIMAL(18,4)	NOT NULL	
Status	NVARCHAR(20)	NOT NULL	Draft/PendingApproval/Approved/Posted/PartiallyPaid/Paid/Cancelled
JournalEntryId	BIGINT	NULL, FK->JournalEntries.Id	Set once posted

UNIQUE (VendorId, VendorInvoiceReference) to stop the same paper bill being entered twice.

PurchaseInvoiceLines

Column	Data Type	Key / Constraint	Description
InvoiceId	BIGINT	FK->PurchaseInvoices.Id	
ItemId	BIGINT	NULL, FK->Items.Id	NULL for a pure expense line
Description	NVARCHAR(300)	NOT NULL	
Quantity / UnitPrice	DECIMAL(18,4)	NOT NULL	
TaxCodeId	BIGINT	NULL, FK	
AccountId	BIGINT	FK->ChartOfAccounts.Id	Expense/asset/inventory account to debit
CostCenterId / ProjectId	BIGINT	NULL, FK	
LineTotal	DECIMAL(18,4)	NOT NULL	

VendorPayments / VendorPaymentAllocations

Money paid out and how it settles specific bills (supports partial/combined payments).

Column	Data Type	Key / Constraint	Description
VendorPayments.VendorId	BIGINT	FK->Vendors.Id	
VendorPayments.Payment Number/PaymentDate	-	NOT NULL	
VendorPayments.Payment Method	NVARCHAR(20)	NOT NULL	Cash/Cheque/BankTransfer/Card
VendorPayments.BankAccountId	BIGINT	FK->BankAccounts.Id	
VendorPayments.Amount / JournalEntryId	-	-	
VendorPaymentAllocations.PaymentId/InvoiceId	BIGINT	FK	Composite unique
VendorPaymentAllocations.AllocatedAmount	DECIMAL(18,4)	NOT NULL	Sum of allocations <= payment amount

VendorCreditNotes / WithholdingTaxEntries

Debit-memos from vendors, and WHT deducted at source.

Column	Data Type	Key / Constraint	Description
VendorCreditNotes.CreditNoteNumber/Date/Amount/Reason	-	NOT NULL	
VendorCreditNotes.JournalEntryId	BIGINT	FK	
WithholdingTaxEntries.SourceDocumentId	BIGINT	NOT NULL	
WithholdingTaxEntries.TaxCodeId/Amount	-	NOT NULL	
WithholdingTaxEntries.CertificateNumber	NVARCHAR(50)	NULL	Issued to vendor as proof of deduction

7. Accounts Receivable (Sales & Customer Collections)

Mirrors Accounts Payable on the sales side: customer master, sales orders, invoices, receipts and aging.

Customers

Customer master, doubles as the AR sub-ledger key.

Column	Data Type	Key / Constraint	Description
CustomerCode	NVARCHAR(20)	UNIQUE per Company	
Name / LegalName	NVARCHAR(200)	NOT NULL	
TaxIdentificationNumber	NVARCHAR(100)	NULL	
CountryCode	CHAR(2)	FK->Countries.Code	
DefaultCurrencyId	CHAR(3)	FK->Currencies.Code	
PaymentTermsDays	INT	NOT NULL DEFAULT 30	
ControlAccountId	BIGINT	FK->ChartOfAccounts.Id	AR control account this customer rolls up to
CreditLimit	DECIMAL(18,4)	NULL	
IsActive	BIT	NOT NULL DEFAULT 1	

CustomerContacts / CustomerBankAccounts

Supporting contact & banking details (structure mirrors Vendor equivalents).

Column	Data Type	Key / Constraint	Description
CustomerContacts.Name/Designation/Email/Phone/IsPrimary	-	-	
CustomerBankAccounts.BankName/AccountNumber/IBAN	-	-	For refunds via bank transfer

SalesOrders / SalesOrderLines

Optional order stage before invoicing.

Column	Data Type	Key / Constraint	Description
SalesOrders.CustomerId/SONumber/OrderDate/Status	-	NOT NULL	
SalesOrderLines.ItemId/Description/Quantity/UnitPrice/TaxCodeId	-	-	

SalesInvoices

Customer invoices - core AR transaction, always posts a balanced JournalEntry.

Column	Data Type	Key / Constraint	Description
BranchId / CustomerId / SalesOrderId	BIGINT	FK	
InvoiceNumber	NVARCHAR(30)	UNIQUE per Company	
InvoiceDate / DueDate	DATE	NOT NULL	
CurrencyId / ExchangeRateToBase	CHAR(3)/DECIMAL	NOT NULL	
SubTotal / TaxTotal / TotalAmount	DECIMAL(18,4)	NOT NULL	
Status	NVARCHAR(20)	NOT NULL	Draft/Approved/Posted/PartiallyPaid/Paid/Overdue/Cancelled
JournalEntryId	BIGINT	NULL, FK->JournalEntries.Id	

SalesInvoiceLines

Column	Data Type	Key / Constraint	Description
InvoiceId / ItemId	BIGINT	FK	
Description / Quantity / UnitPrice	-	NOT NULL	
TaxCodeId / AccountId	BIGINT	FK	Revenue account to credit
CostCenterId / ProjectId	BIGINT	NULL, FK	
LineTotal	DECIMAL(18,4)	NOT NULL	

CustomerReceipts / CustomerReceiptAllocations

Money received and which invoices it settles.

Column	Data Type	Key / Constraint	Description
CustomerReceipts.CustomerId/ReceiptNumber/ReceiptDate	-	NOT NULL	
CustomerReceipts.PaymentMethod/BankAccountId	-	-	
CustomerReceipts.Amount/JournalEntryId	-	-	
CustomerReceiptAllocations.ReceiptId/InvoiceId/AllocatedAmount	-	-	Supports partial & on-account receipts

CustomerCreditNotes / BadDebtWriteOffs

Returns/adjustments and irrecoverable balances.

Column	Data Type	Key / Constraint	Description
CustomerCreditNotes.CreditNoteNumber/Date/Amount/Reason/JournalEntryId	-	-	
BadDebtWriteOffs.InvoiceId/Amount/Reason/ApprovedBy/JournalEntryId	-	-	Requires approval workflow before posting

8. Banking & Cash Management

Bank/cash books, statement import & reconciliation, and cheque tracking - keeps the GL bank balance provably in sync with real-world bank statements.

BankAccounts

Every bank/cash account the company holds, each linked 1:1 to a GL control account.

Column	Data Type	Key / Constraint	Description
AccountName / BankName	NVARCHAR(150)	NOT NULL	
AccountNumber / IBAN / SwiftCode	NVARCHAR(50)	NULL	
CurrencyId	CHAR(3)	FK->Currencies.Code	
GLAccountId	BIGINT	FK->ChartOfAccounts.Id, UNIQUE	One-to-one with the ledger cash/bank account
OpeningBalance	DECIMAL(18,4)	NOT NULL DEFAULT 0	
IsActive	BIT	NOT NULL DEFAULT 1	

BankTransactions

Every movement through a bank account, matched (or not yet) against the bank's own statement.

Column	Data Type	Key / Constraint	Description
BankAccountId	BIGINT	FK->BankAccounts.Id	
TransactionDate / ValueDate	DATE	NOT NULL	
Description / ReferenceNumber	NVARCHAR(300)/NVARCHAR(60)	NULL	
DebitAmount / CreditAmount	DECIMAL(18,4)	NOT NULL DEFAULT 0	
JournalEntryId	BIGINT	FK->JournalEntries.Id	
ReconciliationStatus	NVARCHAR(15)	NOT NULL DEFAULT 'Unreconciled'	Unreconciled/Matched/Reconciled
ReconciledOnStatementId	BIGINT	NULL, FK->BankReconciliations.Id	

BankStatementImports / BankReconciliations

Imported statement files and the monthly reconciliation run.

Column	Data Type	Key / Constraint	Description
BankStatementImports.BankAccountId/ImportDate/FileName/Status	-	-	Raw imported rows staged before matching
BankReconciliations.ReconciliationDate	DATE	NOT NULL	
BankReconciliations.StatementBalance/BookBalance	DECIMAL(18,4)	NOT NULL	
BankReconciliations.Status/ReconciledBy	-	-	InProgress/Completed

ChequeRegister

Tracks issued and received cheques through their lifecycle.

Column	Data Type	Key / Constraint	Description
BankAccountId	BIGINT	FK->BankAccounts.Id	
ChequeNumber / ChequeDate	NVARCHAR(30)/DATE	NOT NULL	
Direction	NVARCHAR(10)	NOT NULL	Issued / Received
PartyType / PartyId	NVARCHAR(20)/BIGINT	NULL	Vendor / Customer
Amount	DECIMAL(18,4)	NOT NULL	
Status	NVARCHAR(15)	NOT NULL	Issued/Presented/Cleared/Bounced/Cancelled
LinkedPaymentId / LinkedReceiptId	BIGINT	NULL, FK	

PettyCashAccounts / PettyCashTransactions

Imprest-style small cash funds held by custodians.

Column	Data Type	Key / Constraint	Description
PettyCashAccounts.Name/ CustodianUserId/GLAccountId/ImprestLimit	-	-	
PettyCashTransactions.Date/Description/Amount/Type/ JournalEntryId	-	-	Type: Expense / Replenishment

9. Fixed Assets & Depreciation

Full asset lifecycle: acquisition, scheduled depreciation posting, transfers, maintenance and disposal.

AssetCategories

Defaults applied to every asset of that category.

Column	Data Type	Key / Constraint	Description
Name	NVARCHAR(100)	NOT NULL	
DefaultUsefulLifeMonths	INT	NOT NULL	
DefaultDepreciationMethod	NVARCHAR(20)	NOT NULL	StraightLine/DecliningBalance/UnitsOfProduction
AssetAccountId / DepreciationExpenseAccountId / AccumulatedDepreciationAccountId	BIGINT	FK->ChartOfAccounts.Id	The 3 GL accounts every posting touches

FixedAssets

Individual asset register.

Column	Data Type	Key / Constraint	Description
AssetCode / Name	NVARCHAR(30)/NVARCHAR(200)	NOT NULL	
CategoryId	BIGINT	FK->AssetCategories.Id	
PurchaseDate / PurchaseCost / SalvageValue	DATE/DECIMAL	NOT NULL	
UsefulLifeMonths / DepreciationMethod	INT/NVARCHAR	NOT NULL	Overrides category default if set
Location / CustodianEmployeeId	-	NULL	
Status	NVARCHAR(20)	NOT NULL DEFAULT 'Active'	Active/UnderMaintenance/Disposed
AccumulatedDepreciation / NetBookValue	DECIMAL(18,4)	NOT NULL	Kept in sync by DepreciationSchedules postings

DepreciationSchedules

One row per asset per period - generated up-front or period-by-period, posted on run.

Column	Data Type	Key / Constraint	Description
AssetId	BIGINT	FK->FixedAssets.Id	
PeriodDate	DATE	NOT NULL	
DepreciationAmount	DECIMAL(18,4)	NOT NULL	
AccumulatedDepreciation / NetBookValue	DECIMAL(18,4)	NOT NULL	Running totals as of this period
JournalEntryId	BIGINT	NULL, FK	
IsPosted	BIT	NOT NULL DEFAULT 0	

AssetDisposals / AssetTransfers / AssetMaintenanceLogs

End-of-life, movement and upkeep history.

Column	Data Type	Key / Constraint	Description
AssetDisposals.DisposalDate/Proceeds/NetBookValueAtDisposal/GainLossAmount/JournalEntryId	-	-	
AssetTransfers.FromLocation/ToLocation/TransferDate/ApprovedBy	-	-	
AssetMaintenanceLogs.Date/Description/Cost/VendorId	-	-	

10. Budgeting

Plan figures per account/period/dimension, compared to actuals via a reporting view (no separate 'actuals' table needed - actuals are simply the posted JournalEntryLines).

BudgetVersions

A named budget scenario for a fiscal year (e.g. 'FY27 Approved', 'FY27 Revised-Q2').

Column	Data Type	Key / Constraint	Description
FiscalYearId	BIGINT	FK->FiscalYears.Id	
Name	NVARCHAR(100)	NOT NULL	
Status	NVARCHAR(15)	NOT NULL DEFAULT 'Draft'	Draft/Approved/Locked
ApprovedBy / ApprovedAt	BIGINT/DATETIME2	NULL	

BudgetLines

Planned amount per account, per dimension, per period.

Column	Data Type	Key / Constraint	Description
BudgetVersionId	BIGINT	FK->BudgetVersions.Id	
AccountId	BIGINT	FK->ChartOfAccounts.Id	
CostCenterId / ProjectId / DepartmentId	BIGINT	NULL, FK	
AccountingPeriodId	BIGINT	FK->AccountingPeriods.Id	
BudgetedAmount	DECIMAL(18,4)	NOT NULL	

UNIQUE (BudgetVersionId, AccountId, CostCenterId, ProjectId, AccountingPeriodId).

11. Basic Inventory / Stock (for COGS costing)

A lean inventory layer sufficient to value stock and post Cost-of-Goods-Sold correctly. If the business needs a full warehouse/logistics system, this module is the integration point (StockTransactions is the seam).

ItemCategories / Items

Product/service master.

Column	Data Type	Key / Constraint	Description
ItemCategories.Name	NVARCHAR(100)	NOT NULL	
Items.ItemCode / Name	NVARCHAR(30)/NVARCHAR(200)	NOT NULL	
Items.ItemType	NVARCHAR(15)	NOT NULL	Inventory / Service / NonInventory
Items.UnitOfMeasure	NVARCHAR(20)	NOT NULL	
Items.CategoryId	BIGINT	FK	
Items.SalesAccountId / PurchaseAccountId / InventoryAssetAccountId	BIGINT	FK->ChartOfAccounts.Id	
Items.CostingMethod	NVARCHAR(15)	NOT NULL	FIFO / WeightedAverage / Standard
Items.StandardCost	DECIMAL(18,4)	NULL	
Items.IsActive	BIT	NOT NULL DEFAULT 1	

Warehouses / StockLevels

Physical locations and current quantities.

Column	Data Type	Key / Constraint	Description
Warehouses.BranchId/Name/Code	-	-	
StockLevels.ItemId/WarehouseId	BIGINT	FK, composite unique	
StockLevels.QuantityOnHand / AverageCost	DECIMAL(18,4)	NOT NULL	Denormalised for fast lookup; rebuilt from StockTransactions

StockTransactions / StockValuationLayers

Immutable movement ledger and FIFO cost layers.

Column	Data Type	Key / Constraint	Description
StockTransactions.ItemId/WarehouseId/TransactionDate	-	NOT NULL	
StockTransactions.TransactionType	NVARCHAR(20)	NOT NULL	PurchaseReceipt/SalesIssue/AdjustmentIn/AdjustmentOut/TransferIn/TransferOut
StockTransactions.Quantity / UnitCost	DECIMAL(18,4)	NOT NULL	
StockTransactions.SourceDocumentType/SourceDocumentId/JournalEntryId	-	-	
StockValuationLayers.ReceiptDate/QuantityRemaining/UnitCost	-	-	One row per FIFO receipt layer, consumed on issue

12. Payroll (Optional Module)

Minimal payroll sufficient to post salary expense/liability to the GL. Statutory calculations (income tax bands, social security) vary heavily by country and should be pluggable per TaxJurisdiction.

Employees

HR-lite master, linked into JournalEntry lines as PartyType='Employee'.

Column	Data Type	Key / Constraint	Description
EmployeeCode / Name	NVARCHAR(30)/NVARCHAR(150)	NOT NULL	
DepartmentId	BIGINT	FK->Departments.Id	
JoinDate	DATE	NOT NULL	
BankAccountNumber / BankName	NVARCHAR(50)	NULL	
BasicSalary	DECIMAL(18,4)	NOT NULL	
PayableAccountId	BIGINT	FK->ChartOfAccounts.Id	Employee-specific liability roll-up
IsActive	BIT	NOT NULL DEFAULT 1	

SalaryComponents / EmployeeSalaryStructures

Configurable earnings/deductions per employee.

Column	Data Type	Key / Constraint	Description
SalaryComponents.Name/Type/IsTaxable/GLAccountId	-	-	Type: Earning / Deduction
EmployeeSalaryStructures.EmployeeId/ComponentId	BIGINT	FK	
EmployeeSalaryStructures.Amount / Percentage / EffectiveFrom	-	-	

PayrollRuns / PayrollTransactions

Monthly payroll processing batch and resulting lines.

Column	Data Type	Key / Constraint	Description
PayrollRuns.AccountingPeriodId/RunDate/Status/JournalEntryId	-	-	Status: Draft/Approved/Posted
PayrollTransactions.PayrollRunId/EmployeeId/ComponentId/Amount	-	-	

EmployeeLoans

Staff advances/loans recovered via payroll deduction.

Column	Data Type	Key / Constraint	Description
EmployeeId / LoanAmount / InstallmentAmount	-	NOT NULL	
StartDate / OutstandingBalance	-	NOT NULL	

13. Financial Reporting & Multi-Company Consolidation

Maps the Chart of Accounts to standard financial statement lines so Balance Sheet / P&L; / Cash Flow / Trial Balance reports can be generated and re-arranged without touching code, plus optional group consolidation.

FinancialStatementTemplates / Lines

Configurable report layout.

Column	Data Type	Key / Constraint	Description
FinancialStatementTemplates.Name/Type/IsDefault	-	-	Type: BalanceSheet/IncomeStatement/CashFlow/TrialBalance
FinancialStatementLines.TemplateId/ParentLineId	BIGINT	FK, self-referencing	Header/sub-total hierarchy
FinancialStatementLines.LineLabel/LineOrder/LineType	-	-	LineType: Header/Detail/Total/Calculation
FinancialStatementLines.Formula	NVARCHAR(300)	NULL	For calculated lines, e.g. GrossProfit = Revenue - COGS

FinancialStatementLineAccounts

Which GL accounts (or account groups) feed each report line.

Column	Data Type	Key / Constraint	Description
StatementLineId	BIGINT	FK->FinancialStatementLines.Id	
AccountId / AccountGroupId	BIGINT	NULL, FK	One of the two is set
SignMultiplier	SMALLINT	NOT NULL DEFAULT 1	+1 or -1, to flip presentation sign

ConsolidationMappings

For groups running several Companies (subsidiaries) that need combined statements.

Column	Data Type	Key / Constraint	Description
SubsidiaryCompanyId / ParentCompanyId	BIGINT	FK->Companies.Id	
OwnershipPercentage	DECIMAL(5,2)	NOT NULL DEFAULT 100	
EliminationAccountId	BIGINT	NULL, FK	For intercompany elimination entries

14. Notifications & Messaging Queue

Supporting infrastructure for approval alerts, due-date reminders and outbound email/SMS.

Notifications

In-app notification feed per user.

Column	Data Type	Key / Constraint	Description
UserId	BIGINT	FK->Users.Id	
Title / Message	NVARCHAR(150)/NVARCHAR(500)	NOT NULL	
Type	NVARCHAR(30)	NOT NULL	ApprovalRequired/InvoiceDue/PaymentReceived/System
LinkUrl	NVARCHAR(300)	NULL	Deep link back to the relevant record
IsRead	BIT	NOT NULL DEFAULT 0	

EmailQueue / SmsQueue

Outbound message queue processed by a background worker (never send synchronously in a web request).

Column	Data Type	Key / Constraint	Description
ToAddress / Subject / Body	NVARCHAR	NOT NULL	
Status	NVARCHAR(15)	NOT NULL DEFAULT 'Pending'	Pending/Sent/Failed
RetryCount / LastError	-	-	
SentAt	DATETIME2	NULL	

Security Architecture Summary

- **Password storage:** Argon2id (preferred) or bcrypt with a unique per-user salt; never MD5/SHA1/plain SHA256.
- **Authentication:** short-lived JWT access tokens (10-15 min) + rotating refresh tokens stored hashed (RefreshTokens table); revoke on logout/password change; detect refresh-token reuse.
- **Authorization:** RBAC via Roles/Permissions/RolePermissions, checked on every API endpoint server-side (never trust a hidden UI element as the only control).
- **Tenant isolation:** CompanyId filter enforced globally (EF Core query filter) plus a server-side check that the authenticated user has UserCompanyAccess to the CompanyId in every request.
- **SQL injection:** use parameterised queries / EF Core LINQ exclusively - never string-concatenate SQL.
- **Transport & storage encryption:** TLS 1.2+ everywhere; encrypt highly sensitive columns (bank account numbers, TIN, 2FA secret) at rest using SQL Server Always Encrypted or application-level AES.
- **Audit trail:** AuditLogs written in the same transaction as the business change; the application's SQL login should only have INSERT rights on this table (no UPDATE/DELETE) so history cannot be erased.
- **Segregation of duties:** approval workflow (Module 2) blocks the same user from both creating and approving high-value documents.
- **Rate limiting & lockout:** throttle the login endpoint; lock the account after N failed attempts (Users.FailedLoginAttempts / IsLocked) with exponential backoff.
- **Least privilege database account:** the application's SQL login should not own DDL rights in production; migrations run under a separate elevated account.
- **Backups & recovery:** automated backups with point-in-time restore; test restores regularly - a backup you have not restored is not a backup.
- **Data retention/GDPR-style rules:** soft-delete + audit trail lets you honour 'right to be forgotten' requests by anonymising personal fields while keeping the financial ledger intact.

Recommended Indexes & Integrity Rules

- Non-clustered index on every foreign key column, especially CompanyId + Date columns (JournalEntries.VoucherDate, PurchaseInvoices.InvoiceDate, SalesInvoices.InvoiceDate).
- Composite index (CompanyId, Status) on every document table used for worklists/dashboards.
- UNIQUE index on (CompanyId, AccountCode), (CompanyId, VendorCode), (CompanyId, CustomerCode), (CompanyId, VoucherNumber), (CompanyId, InvoiceNumber).
- CHECK constraint: JournalEntryLines.DebitAmount >= 0 AND CreditAmount >= 0 AND NOT (DebitAmount > 0 AND CreditAmount > 0) - a line is either a debit or a credit, never both.
- CHECK constraint on ExchangeRates.Rate > 0.
- Trigger or application guard preventing any INSERT/UPDATE/DELETE on JournalEntryLines when the parent JournalEntries.Status = 'Posted' or the AccountingPeriod is closed.
- Filtered index WHERE IsDeleted = 0 on large master tables (Items, Customers, Vendors) to keep active-record lookups fast as history grows.